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Bayesian approach for safety barrier portfolio optimization

A. Mancuso
Department of Mathematics and System Analysis, Aalto University, Finland
Dipartimento di Energia, Politecnico di Milano, Italy

M. Compare
Dipartimento di Energia, Politecnico di Milano, Italy
Aramis s.r.l., Milano, Italy

A. Salo
Department of Mathematics and System Analysis, Aalto University, Finland

E. Zio
Dipartimento di Energia, Politecnico di Milano, Italy
Aramis s.r.l., Milano, Italy
Chair on Systems Science and Energetic Challenge, Fondation EDF (Electricite de France), CentraleSupelec, France

ABSTRACT: The selection and positioning of safety barriers to improve system safety is a fundamental concern in many industrial sectors (e.g., nuclear, process, railway among others). To address it, risk analysts often rely on Bow Tie diagrams, which help model the system accident scenarios and describe the effects of safety barriers that prevent and mitigate the associated risks. Earlier approaches to safety barrier selection and positioning are mainly based on what-if analyses, through which the analysts, based on their experience, add barriers throughout the Bow Tie diagram until the risk of severe accident decreases beyond the accepted threshold. However, the resulting set (i.e., portfolio) of installed barriers may be not cost-efficient. To overcome this limitation, we frame the barrier selection issue within the Portfolio Decision Analysis: the goal is to support the decision maker in the task of identifying the portfolios of safety barriers for a given Bow Tie diagram, which are optimal with respect to system residual risk, reliability and investment costs, and moreover accounting for budget limitations and barrier feasibility constraints. The optimization algorithm used to find the optimal portfolios is based on implicit enumeration, whereby the computational burden remain limited when the number of alternative safety barriers is limited. An illustrative example on the prevention and mitigation of accidental gas release in a process plant is presented to illustrate the method and to outline its possible applications.

1 INTRODUCTION

In industry, the importance of managing assets while achieving minimum risk to human beings, environment and asset integrity asks risk analysts to tackle the issue of selecting and positioning safety barriers to guarantee minimum requirements for asset availability and safety (Hassan and Khan 2012). Currently, many approaches rely on Bow Tie diagrams (Markowski and Kotynia, 2011) to model the accident scenarios from their causes up to the final effects. By what-if analyses, experts include barriers in the system until the risk of critical events decreases beyond the acceptable threshold. These approaches may lead to identify non-efficient sets of safety barriers (Salo et al., 2011).

In this paper, we frame the selection of safety barriers as a Portfolio Decision Analysis problem (PDA, Salo et al. 2011): the decision maker has to make an informed multiple barrier selection (i.e., identify a portfolio) from a discrete set of alternatives (i.e., the barriers) with different features (e.g., prevention, mitigation, reliability, cost, etc.), which is optimal with respect to some criteria (e.g., residual risk, cost, etc.) and fulfil the given constraints on budget and risk acceptance requirements. In this PDA setting, Bow Tie becomes the mathematical model to be embedded in a portfolio optimization algorithm that searches for the optimal barrier portfolios. To the authors' best knowledge, this approach to designing and choosing protective barriers is novel.

To better formalize the qualitative information provided by experts and model the sequence of events with their interdependencies, the methodology proposed relies on Bayesian Belief Networks (BBN). That is, we assume that the Bow Tie usually provided by the risk analysts is transformed into a BBN by using the

approach by (Khakzad et al., 2013). This assumption allows us to focus on developing a methodology to address realistic safety barrier portfolio selection problems within the PDA framework, based on a sound and flexible mathematical model.

The rest of the paper is structured as follows. Section 2 details the methodology, focusing on the graphical representation and the optimization model. Section 3 presents an illustrative example to clarify the proposed methodology. Finally, Section 4 concludes the paper and outlines extensions for future research.

2 PROBLEM FORMULATION

We assume that the Bow Tie diagram provided by the risk analysts has already been converted into a BBN. Formally, a BBN is a directed acyclic graph made up of:

- Nodes $V = \{1, \ldots, N\}$, which are represented by circles and indicate the events composing the accident scenarios (a set of consecutive events which lead to an accident). Among these nodes, the analysis accounts for one or some critical nodes $t \in V^C \subseteq V$, whose outcomes can be the event criticality. These nodes are represented by rounded squares to distinguish them from the non critical ones.
- Directed arcs, $E \subseteq \{(i,j) | i,j \in V,\ i \neq j\}$, which indicate conditional dependencies among nodes. Specifically, arc $(j,i) \in E$ connects node $j \in V$ to node $i \in V$ and indicates that node i is conditional dependent on node j.

The immediate follower nodes of $i \in V$ are indicated by set $V_+^i = \{j | (i,j) \in E\}$, whereas its immediate predecessors are the nodes belonging to set $V_-^i = \{j | (j,i) \in E\}$. On this basis, we can define both the set of leaf nodes $V^L = \{i \in V | V_-^i = \emptyset\}$ and its complement set of dependent nodes $V^D = V \backslash V^L = \{i \in V | V_-^i \neq \emptyset\}$.

A path is a sequence of nodes $(i_1, i_2, \ldots, i_k)$, $k > 1$, such that $(i_j, i_{j+1}) \in E$, $j < k$. The assumed acyclic property of the BBN means that there is no path defined by nodes $(i_1, i_2, \ldots, i_k)$, $k > 1$ such that $(i_j, i_{j+1}) \in E, j < k$ and $i_1 = i_k$.

For every node $i \in V$, we can recursively calculate its depth in the network by:

$$d^i = \begin{cases} 0 & V_-^i = \emptyset \\ 1 + max_{j \in V_-^i} d^j & V_-^i \neq \emptyset \end{cases} \quad (1)$$

The BBN is integrated with decision nodes, which are represented by squares crossing the arcs. Namely, in case of a dependent node (i.e. $d^i \neq 0$), a decision node crossing an arc directed to node i indicates the decision about the action that can be pursued to reduce the risk of the event at node $i \in V$. The conditional probability distribution at node $i \in V$ depends on the action taken at its decision node. In some cases, decisions affect more conditional dependencies; to model this situation, we assume that the square nodes cross all and only the arc(s) representing those conditional dependencies.

On the other hand, the decision on whether and which barrier is to be installed at $i \in V^L$ (i.e., at a leaf node) is represented by a circle inscribed in a square to indicate that node $i \in V^L$ is combined with the decision node impacting on it. In keeping general, from now on these decisions about barrier selection are referred to as actions (Salo et al., 2011).

Formally, at node $i \in V$, the set of alternative applicable actions is $\mathcal{A}^i = \{1, \ldots, |\mathcal{A}^i|\}$, where $|\cdot|$ indicates the cardinality of the set. In general, these actions have different characteristics, depending for example on whether they are prevention or mitigation barriers.

The decision on whether and which action has to be applied at node $i \in V$ is indicated by the binary decision variable z_a^i, which is set to 1 if $a \in \mathcal{A}^i$ is included in the action portfolio and to 0, otherwise. This way, an action portfolio $A \subseteq \mathsf{X}_{i \in V} \mathcal{A}^i$ is uniquely defined by the binary vectors $\mathbf{z}^i = [z_a^i], \forall a \in \mathcal{A}^i,\ i \in V$. The term $\mathsf{X}_{i \in V}$ indicates the Cartesian product of sets $\mathcal{A}^i$.

For some nodes $i \in V$, no action may be feasible; this is modelled by $\mathcal{A}^i = \emptyset$ and, thus, $|\mathcal{A}^i| = 0$. Yet, we assume that actions at node $i \in V$ are mutually exclusive, meaning that a single action (or eventually no action) can be selected from set $\mathcal{A}^i$. This constraint is formalized by:

$$\sum_{a \in \mathcal{A}^i} z_a^i \leq 1 \quad \forall i \in V \quad (2)$$

For clarity, in Figure 1, we show an example of a BBN, where $V^L = \{1,2\}$, $V^D = \{3,4,5,6,7\}$, the critical node $t = 7$ and actions can be applied to each of these nodes 1, 3, 4 and 6. If we assume that there are three possible actions at each of these nodes, the action portfolio $A = \{a_2^1, a_1^3, a_3^4, a_2^6\}$, where the superscript and the subscript indicate the node and the action index respectively, is defined by the binary vectors:

$$\mathbf{z}^1 = [0,1,0]$$

$$\mathbf{z}^3 = [1,0,0]$$

$$\mathbf{z}^4 = [0,0,1]$$

$$\mathbf{z}^6 = [0,1,0]$$

2.1 *Optimization model*

Once an action is introduced into a system, its influence on the accident scenario evolution depends on the magnitude of deviations (e.g., intensity of a gas release) and on the effectiveness of the actions in counteracting the accident. In industrial practice, such extent is often described in terms of a multi-state random variable.

Let X^i be the random variable representing the uncertainty in the state of node $i \in V$ (i.e., the extent of the deviation). The realization of X^i belongs to the set of states $\mathcal{S}^i = \{0, \ldots, |\mathcal{S}^i|\}$, where state 0 indicates

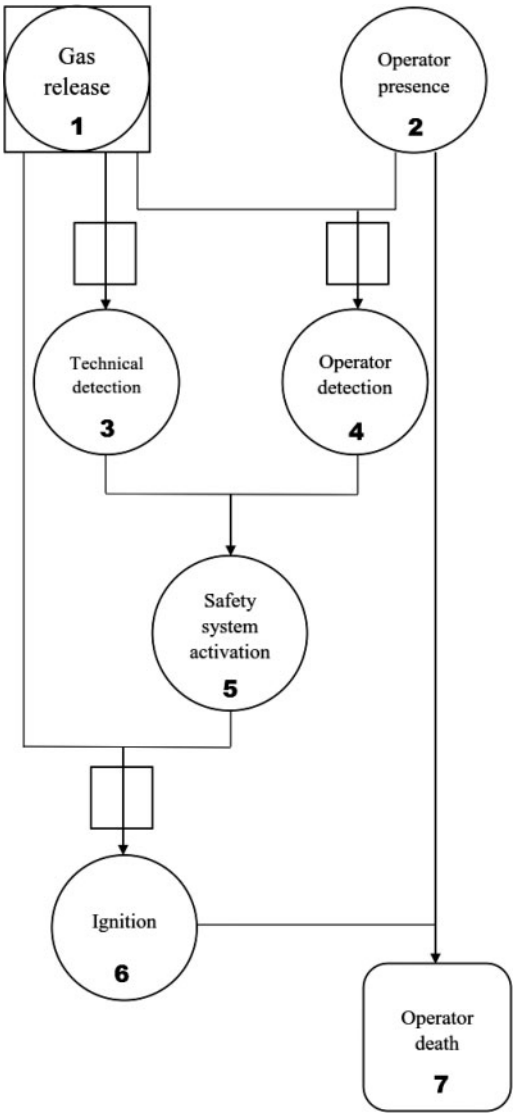


Figure 1. BBN for flammable gas release.

that the event at node $i \in V$ does not occur, whereas $|\mathcal{S}^i| \geq 1$ refers to the highest magnitude of such event.

The uncertainty in the realization of X^i, $i \in V^L$ is described by the probability mass distribution $\mathcal{P}_{X^i}(s) = p(X^i = s) \geq 0$ of node $i \in V^L$ being in state $s \in \mathcal{S}^i$, given that

$$\sum_{s \in \mathcal{S}^i} \mathcal{P}_{X^i}(s) = 1, \ \forall i \in V^L. \tag{3}$$

The application of action $a \in \mathcal{A}^i$ to leaf node $i \in V^L$ changes $\mathcal{P}_{X^i}(s)$ into $\mathcal{P}_{X^i_a}(s)$, where

$$\sum_{s \in \mathcal{S}^i} \mathcal{P}_{X^i_a}(s) = 1, \quad \forall a \in \mathcal{A}^i \tag{4}$$

and the total probability that node $i \in V^L$ is in state $s \in \mathcal{S}^i$ is

$$\mathcal{Q}_{X^i}(s) = \sum_{a \in \mathcal{A}^i} z^i_a \, \mathcal{P}_{X^i_a}(s). \tag{5}$$

Note that for every node the way an action modifies the probability distribution representing the deviation from state $s = 0$ changes case by case. For this reason, the rules to get the posterior probability need to be elicited from experts.

At every dependent node $i \in V^D$, the probability of being in state $s \in \mathcal{S}^i$ depends on the states of its predecessor random variables. To model this relationship, we define $\mathbf{X}^i_-$ as the $|V^i_-|$-dimensional vector composed by the random variables X^j, $\forall j \in V^i_-$.

Let $\mathcal{S}^i_-$ be the set of the Cartesian product of all the set of states $\mathcal{S}^j$, $j \in V^j_-$. Then, a possible realization of $\mathbf{X}^i_-$ is indicated by vector $\mathbf{x}^i \in \mathcal{S}^i_-$, where its j-th entry x^i_j denotes the realization of the corresponding random variable $X^j, j \in V^i_-$.

In this setting, the conditional probability of state $s \in \mathcal{S}^i$ at node $i \in V^D$, given $\mathbf{x}^i \in \mathcal{S}^i_-$, is

$$\mathcal{Q}_{X^i|\mathbf{x}^i}(s) = \sum_{a \in \mathcal{A}^i} z^i_a \, \mathcal{P}_{X^i_a|\mathbf{x}^i}(s) \tag{6}$$

where $\mathcal{P}_{X^i_a|\mathbf{x}^i}(s)$ is the conditional probability of state $s \in \mathcal{S}^i$ at node $i \in V^D$, given the realization $\mathbf{x}^i$ of its predecessors with their corresponding actions, and the selection of action $a \in \mathcal{A}^i$.

From above, the total probability of state $s \in \mathcal{S}^i$ at node $i \in V^D$ can be written as

$$\mathcal{Q}_{X^i}(s) = \sum_{\mathbf{x}^i \in \mathcal{S}^i_-} \left[\sum_{a \in \mathcal{A}^i} z^i_a \, \mathcal{P}_{X^i_a|\mathbf{x}^i}(s) \right] \prod_{j \in V^i_-} \mathcal{Q}_{X^j}(x^i_j) \tag{7}$$

where the first operator sum considers all the possible realizations $\mathbf{x}^i \in \mathcal{S}^i_-$.

In turn, the total probability $\mathcal{Q}_{X^i}(s)$ is a multiplicative function of the actions that have been applied along the paths leading from the leaf nodes to $i \in V^D$.

As mentioned before, the BBN contains a critical value node $t \in V^C$. The expected utility function assigned to t is

$$\mathcal{U}^t(A) = \sum_{s \in \mathcal{S}^t} \mathcal{Q}_{X^t}(s) \, u^t(s) \tag{8}$$

where $u^t(s)$ quantifies the criticality level of state $s \in \mathcal{S}^t$ defined by the DM. Namely, $u^t(s) = 0$ if state $s \in \mathcal{S}^t$ is not critical according to the DM judgements, otherwise $u^t(s) > 0$.

An estimate for $u^t(s)$, $\forall s \in \mathcal{S}^t$ can be derived through a trade-off weighting approach, such as SMART (Edwards, 1977), SWING (von Winterfeldt & Edwards, 1986) or SMARTS (Edwards & Barron, 1994) where the only alternative is the critical node

$t \in V^C$ and the attributes are the states $s \in \mathcal{S}^t$. If the critical node $t \in V^C$ is described by binary states, then the goal could be to minimize the total probability $\mathcal{Q}_{X^t}(s)$ of the critical node by setting $u^t(0) = 0$ and $u^t(1) = 1$.

Finally, different actions $a \in \mathcal{A}^i$ have different costs C_a; then, the optimization model accounts for the overall cost of the action portfolio, which must not exceed the available budget B.

Based on the considerations above, the problem of selection of safety actions is formalized as the following portfolio optimization problem

$$\min_{A \subseteq \mathsf{X}_{i \in V} \mathcal{A}^i} \quad \mathcal{U}^t(A) \tag{9}$$

$$\mathcal{Q}_{X^i}(s) = \sum_{a \in \mathcal{A}^i} z_a^i \, \mathcal{P}_{X_a^i}(s) \quad \forall i \in V^L \tag{10}$$

$$\mathcal{Q}_{X^i}(s) = \sum_{\mathbf{x}_-^i \in \mathcal{S}_-^i} \left[\sum_{a \in \mathcal{A}^i} z_a^i \, \mathcal{P}_{X_a^i | \mathbf{x}^i}(s) \right] \prod_{j \in V_-^i} \mathcal{Q}_{X^j}(x_j^i)$$

$$\forall i \in V^D \tag{11}$$

subject to the constraints

$$\sum_{a \in \mathcal{A}^i} z_a^i \leq 1, \quad \forall i \in V \tag{12}$$

$$\sum_{i \in V} \sum_{a \in \mathcal{A}^i} z_a^i \, C_a \leq B \tag{13}$$

$$\mathbf{z}^i \in \{0,1\}^{|\mathcal{A}^i|} \quad \forall i \in V \tag{14}$$

The calculation of the total probabilities $\mathcal{Q}_{X^i}(s)$ starts from the leaf nodes $i \in V^L$ and, then, proceeds up to the total probabilities of dependent nodes $i \in V^D$ by increasing the node depth d^i. This is necessary because the calculation of the total probability $\mathcal{Q}_{X^i}(s)$ requires the total probabilities $\mathcal{Q}_{X^j}(s)$ of all the predecessors $j \in V_-^i$.

To identify the optimal portfolios of safety actions, we employ the implicit enumeration algorithm derived from Liesiö (2014) and reported in Appendix. The algorithm is computationally efficient, although the computational time depends on the number of nodes of the system and the amount of alternative actions per node.

3 ILLUSTRATIVE EXAMPLE

In a process plant, the accidental release of flammable gas can be detected either by an automatic gas detection system or by a process operator, who is present 30% of the time and can detect gas only if she/he is present in the area where the gas is released. Table 1 shows probabilities of detection, false positive and false negative of both operator and technical system: state 1 refers to the occurrence of the detection event, whereas state 0 refers to its opposite. This way, a failure to detect is represented by states of "Gas release" and "Technical detection" equal to 1 and 0 respectively, whereas a false alarm is represented by states of "Gas release" and "Technical detection" equal to 0 and 1 respectively.

Table 1. Detection probabilities.

		Technical detection	
	State	1	0
Gas release	1	0.99	0.01
	0	0.1	0.9
		Operator detection	
	State	1	0
Gas release	1	0.95	0.05
	0	0.3	0.7

If the gas release is detected, the safety system is activated either automatically or manually by the operator to prevent a larger gas release and the possible consequent ignition. The safety system fails with probability of 4% if the release is detected by the automatic system only, and with probability of 2% if the release is detected by the operator or both.

The released gas is ignited with probability of 10%, but the safety system can sensibly reduce this probability value to 1%. The frequency of gas release is 0.1 events per year. If the operator is in the area at the time of the ignition, the probability of being killed is 20%.

The probability values used in this example are illustrative, whereas Figure 1 shows the related BBN. This is representative of the Event Tree part of the Bow Tie diagram modeling only the flammable gas release accident scenario (Markowski and Kotynia, 2011): the Fault Tree modeling the causes of gas release has been disregarded to keep the illustrative example as simple and clear as possible. This also allows us working only with mitigation barriers, which is a further simplification.

The BBN shows 7 nodes $i \in V$, each of them characterized by $|\mathcal{S}^i| = 2$ states. Specifically, state $S^i = 1$ refers to the occurrence of the event at node $i \in V$, whereas 0 refers to its opposite.

The optimization model aims to minimize the probability of "Operator death", while ensuring that the probability of "Ignition" is smaller than 0.1%. To minimize the probability of the operator death, it is possible to apply one or several safety actions to some selected node. For simplicity, each action is assigned a Performance Factor (PF), which represents the relative reduction brought by the action on the probability of having the event which it impacts on: the smaller the PF, the better the safety action, the larger its cost. For example, for $i = 1$ a generic action $a \in \mathcal{A}^1$ would modify the probability of "Gas release" according to the following equations:

$$P_{X_a^1}(1) = P_{X^1}(1) \, PF_a \tag{15}$$

Table 2. Characteristics of applicable actions.

Index	Action	Cost [€]	PF
Anticorrosion paint	a_1^1	1000	0.8
Pipe coating	a_2^1	2500	0.3
Catalytic	a_1^3	500	0.8
Infrared	a_2^3	800	0.5
Ultrasonic	a_3^3	1500	0.2
Gas odour	a_1^4	400	0.7
Fire sprinkler system	a_1^6	2000	0.4
Hypoxic air technology	a_2^6	4000	0.1
Fire protection synergy	a_3^6	6000	0.02

Table 3. Posterior probabilities for action on node $i = 1$.

	x^1	
	0	1
$\mathcal{P}_{X^1_{a_1^1}}(s)$	0.92	0.08
$\mathcal{P}_{X^1_{a_2^1}}(s)$	0.97	0.03

$$P_{X_a^i}(0) = 1 - P_{X_a^i}(1) \tag{16}$$

In Table 2, we list the alternative safety actions and the relative parameters. For example, three possible actions can be taken at node 6, which are reported in the last row. If we select the first action, i.e., "fire sprinkler system", the probability of ignition of the released gas reduces by 40% and the corresponding cost is €2000. The fire protection synergy refers to a combination of "fire sprinkler system" and "hypoxic air technology": if both systems are installed, there is a synergistic effect which outperforms that of installing both barriers as if they were independent: in this case, the ignition probability is reduced by 0.02 instead of $0.4 \cdot 0.1 = 0.04$. Note that in Table 2 the superscript and the subscript (second column) represent the node and the index of the action, respectively.

Note that for every action $a \in \mathcal{A}^i$ we need to elicit the relative cost and the conditional probabilities given the application of the action.

The posterior probabilities $\mathcal{P}_{X_a^i}(s)$ of the chance nodes impacted by the decision nodes are defined in the following Tables 3–6.

The optimal action portfolios are identified by the implicit enumeration algorithm in Appendix. It is run repeatedly by increasing the budget from €0 to €10400 to cover the full range of investment choices (Liesiö et al., 2008; Liesiö, 2014). Computation of the optimal action portfolios takes around 2 seconds on a personal computer (Intel Core i5-5300U, 2.3 GHz, 8 GB).

The results are shown in Figures 2 and 3. Figure 2 shows the accident probability in case the optimal action portfolio is sought with the budget threshold indicated in abscissa. The larger the available budget, the more effective the barriers that can be installed, the smaller the residual risk of operator death. Note that the constraint on the ignition probability cannot be satisfied if the budget is smaller than €4000. For this, no action is taken when $B \leq$ €4000 (Figure 3).

Table 4. Posterior probabilities for action on node $i = 3$.

$\mathcal{P}_{X^3_{a_1^3}}(s)$	x^3	
x^1	0	1
0	0.72	0.28
1	0.008	0.992
$\mathcal{P}_{X^3_{a_2^3}}(s)$	x^3	
x^1	0	1
0	0.45	0.55
1	0.05	0.95
$\mathcal{P}_{X^3_{a_3^3}}(s)$	x^3	
x^1	0	1
0	0.18	0.82
1	0.02	0.98

Table 5. Posterior probabilities for action on node $i = 4$.

$\mathcal{P}_{X^4_{a_1^4}}(s)$		x^4	
x^1	x^2	0	1
0	0	1	0
	1	0.985	0.015
1	0	1	0
	1	0.21	0.79

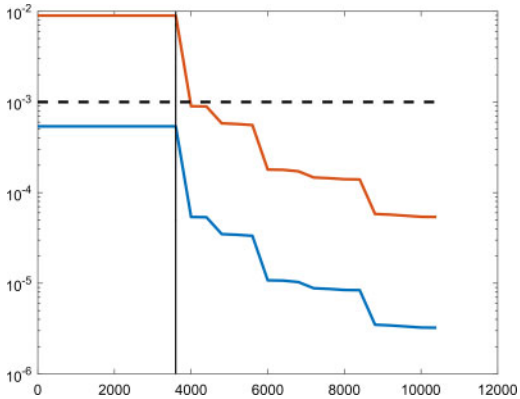


Figure 2. Probability of accident on operator.

Table 6. Posterior probabilities for action on node $i = 6$.

$\mathcal{P}_{X^6_{a^6_1}}(s)$		x^6	
x^1	x^5	0	1
0	0	1	0
	1	1	0
1	0	0.96	0.04
	1	0.996	0.004
$\mathcal{P}_{X^6_{a^6_2}}(s)$		x^6	
x^1	x^5	0	1
0	0	1	0
	1	1	0
1	0	0.99	0.01
	1	0.999	0.001
$\mathcal{P}_{X^6_{a^6_3}}(s)$		x^6	
x^1	x^5	0	1
0	0	1	0
	1	1	0
1	0	0.998	0.002
	1	0.9998	0.0002

In Figure 3, the histograms show the optimal actions for every node, according to the available budget (abscissa).

Given a budget of €4000, the results indicate to apply hypoxic air technology to protect the system from ignition (Figure 3, bottom-right). By increasing the constraint on available budget, the composition of the optimal portfolios changes. In particular, actions to reduce the probability of gas release and to improve the technical detection are the most effective (Figure 3, top-left). As soon as the fire protection synergy becomes feasible, this action is introduced in the optimal portfolios (Figure 3, bottom-right) even though the effort on risk control on the other nodes is reduced. Pipe coating is again adopted when the budget allows applying both the fire protection synergy and the pipe coating.

Note that the gas odour is applied only in case the budget constraint does not allow any other action to be introduced (Figure 3, bottom-left). The effectiveness of the gas odour is very small, as it is useful only if the operator is present.

Note also that the composition of the optimal portfolios heavily depends on the budget, which is a consequence of the fact that the optimization model does not pursue a local optimization (at the single node level); rather, it identifies the optimal action portfolio for the whole system, accounting for budget and logical constraints.

Finally, the illustrative example could be extended to account for the FT that leads to the gas release and

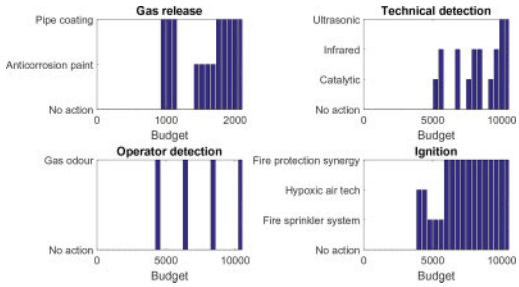


Figure 3. Optimal actions per event.

to describe the nodes through multiple states, instead of binary states only. Moreover, in order to better describe the extent of the harm, the proposed modeling framework allows considering multi-objective functions, i.e., utility functions of several critical nodes.

3.1 *Additional insights*

The definition of the optimal action portfolio accounts for the dependencies between nodes of the system under study. For example, consider an AND gate which connects two initiating events leading to an accident; then, the appropriate action to apply to the critical node could aim not to mitigate the effects, but to modify the dependencies among these two initiating events so that, for example, they cannot occur together (i.e., XOR gate).

This emphasizes one of the advantages of framing the problem of selection of safety actions within the PDA: the model is not focused on local optimization; rather, it identifies the optimal action portfolio for the whole system under analysis.

Another advantage is the possibility to model multiple states for every node. For example, consider node 7 in Figure 1; the expert can model the multiple states of the operator harm as: "No harm", "Minor injury", "Severe injury" and "Operator death". This way, the representation of the system is more realistic. Nevertheless, the expert needs to define the state probabilities according to the identified accident scenarios. This introduces to the main drawback of this methodology, which is the strong effort in elicitation and definition of parameters, which is typical in risk analysis.

Furthermore, the optimization model can be integrated to take a more accurate decision. The DM could introduce additional logical constraints to bound the action selection. For instance, if actions on technical and operator detection in Figure 1 are mutually exclusive, the DM would model the following additional constraint

$$\sum_{a \in \mathcal{A}^3} z_a^3 + \sum_{a \in \mathcal{A}^4} z_a^4 \leq 1. \tag{17}$$

On the other hand, if at least one action among the ones available at nodes 3 and 4 must be applied, then the additional constraint would be

$$\sum_{a\in\mathcal{A}^3} z_a^3 + \sum_{a\in\mathcal{A}^4} z_a^4 \geq 1. \tag{18}$$

In some cases, the same action impacts on different nodes. This action must be included in the sets $\mathcal{A}^i$ of nodes $i \in V$, whose state probabilities are influenced by such action. For instance, if action a' impacts on both nodes 3 and 4 in Figure 1 then $a' \in \mathcal{A}^3, a' \in \mathcal{A}^4$ and the model would account for the following additional constraint

$$z_{a'}^3 + z_{a'}^4 = 0 \tag{19}$$

These constraints would change the optimal portfolios of safety actions, together with many others that can be modelled in PDA according to the needs and the specific features of the system.

Finally, we present two different approaches to cope with the case of multiple critical nodes $t \in V^C$ such as "Operator death" and "System Unavailability". First, the DM can introduce additional logical constraints so that the total probabilities $\mathcal{Q}_{X^t}(s)$ of state $s \in \mathcal{S}^t$ must not exceed a given threshold $\epsilon^t(s)$ such that

$$\mathcal{Q}_{X^t}(s) \leq \epsilon^t(s), \quad t \in V^C \tag{20}$$

The constraints need to be fulfilled for the risk of the system under study to be acceptable.

The alternative way is a multi-objective optimization model which accounts for the expected utilities $\mathcal{U}^t$ of critical nodes $t \in V^C$. This way, the optimal action portfolios would be selected among the Pareto optimal frontier, i.e. the set of action portfolios that are not dominated (Liesiö, 2008). Specifically, let t_1, $t_2 \in V^C$ be two critical events whose impacts are described by the utility functions $\mathcal{U}^{t_1}$ and $\mathcal{U}^{t_2}$, respectively. Given that they are both critical events, it is usually the case that no preference structure between the events can be identified; thus, the analysis leads to the identification of the Pareto optimal frontier, whose dominance condition between two action portfolios A' and A is given by

$$A' \succ A \Leftrightarrow \begin{cases} \mathcal{U}^{t_1}(A') \leq \mathcal{U}^{t_1}(A) \ \wedge \ \mathcal{U}^{t_2}(A') < \mathcal{U}^{t_2}(A) \\ \mathcal{U}^{t_1}(A') < \mathcal{U}^{t_1}(A) \ \wedge \ \mathcal{U}^{t_2}(A') \leq \mathcal{U}^{t_2}(A) \end{cases}$$

In conclusion, several ways to customize the optimization model described in subsection 2.1 are possible according to the specific system under study. This issue will be investigated in future research work.

4 CONCLUSION AND FUTURE RESEARCH

In this paper, we have developed a methodology to support the selection of cost-efficient portfolios of safety actions in high-risk installations. The problem has been framed within the PDA framework to help the DM choose action portfolios that efficiently improve the system safety. The feasibility of the method has been illustrated through an example concerning an accident scenario of flammable gas release in a process plant.

Topics for future research include the following.

Our methodological framework assumes that the information required to set the conditional distributions can be obtained. This poses two main issues. On one side, the optimization model must be able to account for the imprecision usually affecting incomplete datasets and qualitative statements provided by experts. For example, the expert may provide imprecise values of PF, and the costs of the actions may be imprecise as well. Such imprecision must be properly represented and propagated through the model, so that final decisions account for that. On the other hand, a method to facilitate the elicitation needs to be developed, to avoid asking experts to answer many and complex questions.

Furthermore, in future work we intend to integrate the developed approach into a modeling framework to mitigate the post-decision disappointments (Vilkkumaa et al., 2014). In fact, there may be a gap between the "ex-ante" risk estimation, which in the proposed framework is based on rough qualitative values, and the outcomes of the deep risk analyses performed "ex-post" on the optimized design. It is important to select the optimal portfolios of safety actions accounting for the possible errors in the risk estimation.

Finally, an additional challenge is to extend the application of the proposed methodology to non-coherent and dynamic systems. In this case, modeling the accidental scenarios and action behaviour becomes more complicated. Techniques from the Integrated Deterministic and Probabilistic Safety Assessment (e.g., Zio 2014) field could be used to address this issue.

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APPENDIX

The algorithm aims at defining the optimal action portfolio $\mathbf{z}_*$ for the objective function $\mathcal{U}^t_* = \mathcal{U}^t(\mathbf{z}_*)$.

Data: $\mathcal{U}^t(\cdot)$, R, $\mathbf{b}$, $\epsilon^i(s)$
Result: $\mathbf{z}_*$, $\mathcal{U}^t_*$
$\mathbf{z} = [0, ..., 0]^T$, $k \leftarrow 1$, $\mathbf{z}_* \leftarrow \emptyset$, $\mathcal{U}^t_* \leftarrow \infty$;
if $\mathbf{z} \in Z_F$ **then**
  $\mathbf{z}_* \leftarrow \mathbf{z}$, $\mathcal{U}^t_* \leftarrow \mathcal{U}^t(\mathbf{z})$;
end
while $k > 0$ **do**
  Forward-loop;
  while $k \leq m$ **do**
    $z_k \leftarrow 1$;
    if $\mathbf{z} \in Z_F$ and $\mathcal{U}^t(\mathbf{z}) < \mathcal{U}^t_*$ and $\mathcal{Q}_{X^i}(s) \leq \epsilon^i(s)$, $s \in \mathcal{S}^i$, $i \in V$ **then**
      $\mathbf{z}_* \leftarrow \mathbf{z}$, $\mathcal{U}^t_* \leftarrow \mathcal{U}^t(\mathbf{z})$;
    end
    if $\sum_{j=1}^{k} z_j r^l_j + \sum_{j=k+1}^{m} \min\{0, r^l_j\} > b^l \; \forall l = 1, ..., \mathcal{L}$ **then**
      break loop
    end
    $k \leftarrow k + 1$;
  end
  Backtrack-step;
  $z_m \leftarrow 0$;
  $k \leftarrow \max(\{j | z_j = 1\} \cup \{0\})$;
  if $k > 0$ **then**
    $z_k \leftarrow 0$;
    $k \leftarrow k + 1$
  end
end

Figure 4. Implicit enumeration algorithm.

In particular, every action portfolio is described by the binary vector $\mathbf{z} = [z_1, \ldots, z_m]$ which is the concatenation of vectors $\mathbf{z}^i$, $\forall i \in V$ such that $\mathbf{z}_{|\mathcal{A}_i|+a} = z^{i+1}_a$, $a = 1, \ldots, |\mathcal{A}^{i+1}|$. The size of the binary vector $\mathbf{z}$ is $m = \sum_{i \in V} |\mathcal{A}^i|$.

The model accounts for the objective function $\mathcal{U}^t(\cdot)$, the budget and logical constraints, and the upper bounds $\epsilon^i(s)$ of the total probabilities of selected critical states $s \in \mathcal{S}^i$, $i \in V$ (**??**).

In particular, the set of feasible action portfolios is defined by a set of linear inequalities whose coefficients are recorded in matrix $R \in \mathbb{R}^{\mathcal{L} \times m}$ ($r^l_j = [R]_{lj}$) and vector $\mathbf{b} = [b^1, \ldots, b^{\mathcal{L}}] \in \mathbb{R}^{\mathcal{L}}$. This includes the budget constraints. The set of feasible action portfolios is

$$Z_F = \{\mathbf{z} \in \{0,1\}^m | R\mathbf{z} \leq \mathbf{b}\} \tag{21}$$

where $\leq$ holds component-wise.

Besides other logical constraints defined by the DM, these constraints include the budget threshold and the constraints related to the uniqueness of the action applied at each node.

In Figure 4, we show the implicit enumeration algorithm aimed to define the optimal action portfolio to minimize the risk of critical events.